

Data Envelopment Analysis Models Connected in Time Series: A Case Study Evaluating COVID-19 Pandemic Management in Some European Countries

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Abstract—Efforts of individual countries fighting the COVID-19 pandemic may significantly vary. In this work, we compare the overall efficiency of COVID-19 anti-pandemic management of selected European countries using data envelopment analysis (DEA) models connected in weekly time series for summer 2021. To evaluate the COVID-19 transmission mitigation of the countries, we consider a proportion of fully vaccinated people, relative numbers of hospitalized patients, and a reproductive number as an indicator of the rate of current regional COVID-19 pandemic transmission. All DEA models, including those with variable returns to scale and free disposability hulls, require no assumptions, unlike regression models. By connecting the models in time series, we try to identify countries with the highest long-term efficiencies that may serve as examples for other countries. Of course, the situation is always more complex than the data envelopment models, even if connected in time series, may describe; thus, the results are rather illustrative and propose the aspect of models connection in time series.

Keywords—COVID-19, COVID-19 transmission mitigation, data envelopment analysis, data envelopment analysis connected in time series, COVID-19 anti-pandemic management.

I. INTRODUCTION

While the vaccination against COVID-19 raised at the beginning of the year 2021 hopes for promising changes in global pandemic developments, the alone vaccination “as-is” seems not to be a “panacea” for COVID-19 defeat [1]. With no following other anti-pandemic and hygienic rules, e. g. frequent hand washing and disinfection, keeping feasible physical distancing and avoiding physical greetings, and wearing a mask, the vaccination itself is not sufficient to control the COVID-19 pandemic [2]. As indirect proof, there are well-vaccinated populations, but they still suffer from COVID-19 infections and public transmission. Moreover, an important role of non-pharmaceutical measurements is confirmed by very up-to-date papers [3], too. Thus, the proportion of fully vaccinated inhabitants is not strictly proportional to a level of COVID-19 pandemic mitigation in a given country [4].

In this paper, we used several variables to compare COVID-19 anti-pandemic overall management efficiency and

COVID-19 pandemic transmission mitigation in selected European countries. We focus on the proportions of fully vaccinated inhabitants, numbers of hospitalized patients per one million inhabitants, and a local pandemic reproductive rate estimated by the reproduction number [1]. Assuming the overall hygienic level and other hygienic and anti-epidemic rules differ between countries, the proportion of fully vaccinated people does not determine how much the COVID-19 pandemic is under control [1]. Thus, we take the proportion of fully vaccinated people as an initial piece of information and evaluate the other variables – the lower the number of hospitalized patients is as well as the lower the reproduction rate is, the better is COVID-19 pandemic situation in a given country under control and, consequently, the higher efficiency the country anti-pandemic management has got.

The efficiency of the anti-pandemic management is estimated using data envelopment analysis (DEA) models [5]. We use several different approaches to DEA models. However, regardless of them, an idea behind the DEA models is common for all (sub)approaches – an estimation of production-possibility frontiers using input variables (e. g., in our case, proportion of fully vaccinated) and output variables (e. g., in our case, number of hospitalized patients and reproduction rate). To be more specific, to estimate how efficient is a given country at a given vaccination level in controlling anti-pandemic policy and motivation of inhabitants to follow other hygienic rules to reduce the number of hospitalized patients and keep the reproduction rate low.

We use weekly observations for the variables as mentioned above, covering a period of summer 2021. In addition, as a novelty feature, we propose connecting estimated efficiencies output by individual DEA models for individual (week) time points into time series to evaluate better which countries can keep a good efficiency of COVID-19 anti-pandemic management in a longer time period.

In comparison to regression models, DEA models are assumptions-free, nonparametric, and generally require fewer entry data [6], which is more than welcome when a source of data is limited (as typical for COVID-19-related data).

Furthermore, DEA models enable us to indicate so-called peer units, i. e. in our case, the countries with optimal technical efficiency that may serve as a “good example” for others in the anti-pandemic management.

II. DATASET AND METHODS

A. Description of used data

Data used in analyses behind the paper are publicly available and come from ourworldindata.org/coronavirus [7]. As mentioned above, for each country, we are interested in (i) a cumulative proportion of fully vaccinated per one hundred inhabitants, (ii) a cumulative number of hospitalized patients per one million inhabitants, and (iii) a reproduction rate measured by the reproduction number.

To map the current COVID-19 anti-pandemic management situation, we use data for the period of summer 2021, i. e., weekly observations from July 15, 2021, to October 15, 2021. We select all European countries with complete observations in the given time period, i. e. 18 countries in total, namely (in alphabetical order; an international country's abbreviation always follows a name) Austria (AUT), Belgium (BEL), Bulgaria (BGR), Czechia (CZE), Denmark (DNK), Estonia (EST), Finland (FIN), France (FRA), Ireland (IRL), Italy (ITA), Latvia (LVA), Malta (MLT), Norway (NOR), Poland (POL), Slovakia (SVK), Slovenia (SVN), Switzerland (CHE), and the United Kingdom (GBR), respectively.

B. Data envelopment analysis (DEA) models

The DEA models were used to estimate the country-based efficiency of anti-pandemic management, measured by variables of mitigation of COVID-19 public transmission. The DEA models enable the calculation of production-possible frontiers using technical efficiencies of decision-making units (DMU). i. e. independent countries. Decision-making units, i. e. countries, are supposed to “work” with an input variable, i. e. the proportion of fully vaccinated people, and by their anti-pandemic policy activities, manage values of output variables, i. e., numbers of hospitalized patients and reproduction rate. The DEA models may be oriented either to inputs, or to outputs. The input (output) orientation means that ineffective units should manage their inputs (outputs) according to effective (peer) ones keeping their outputs (inputs) constant to become effective. Assuming there are n independent decision-making units (DMU), k input variables $\{X_1, X_2, \dots, X_k\}$ and l output variables $\{Y_1, Y_2, \dots, Y_l\}$, we define a criterial matrix C_t for time point t as follows [4],

$$C_t = \begin{pmatrix} x_{1,1,t} & \dots & x_{1,k,t} & y_{1,1,t} & \dots & y_{1,l,t} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ x_{n,1,t} & \dots & x_{n,k,t} & y_{n,1,t} & \dots & y_{n,l,t} \end{pmatrix}, \quad (1)$$

where units are listed in rows, values of inputs and outputs for time point t are listed in columns and marked by $x_{i,j,t}$ and $y_{i,j,t}$, respectively. To estimate the DEA output-oriented model in time point t means calculation of a relative efficiency $\phi_{\mu,t}$ for each unit μ ,

$$\phi_{\mu,t} = \frac{\sum_{i=1}^l v_{\mu,i,t} y_{\mu,i,t}}{\sum_{j=1}^k u_{\mu,j,t} x_{\mu,j,t}}, \quad (2)$$

where $\mu \in \{1, 2, \dots, n\}$ and $u_{\mu,j,t}$, $v_{\mu,i,t}$ are individual weights for inputs and outputs, respectively, in time point t . The efficiencies in output-oriented models belong to interval $\phi_{\mu,t} \in \langle 1, +\infty \rangle$. The input-oriented DEA model uses a reciprocal version of the efficiency, i. e. $\frac{1}{\phi_{\mu,t}}$. The efficiencies of input-oriented models belong to interval $\frac{1}{\phi_{\mu,t}} \in \langle 0, 1 \rangle$. Effective are those units having the efficiency equaled to 1, are called effective (peer) units and may serve as examples for ineffective ones; the weights $u_{\mu,j,t}$, $v_{\mu,i,t}$ reveal in what proportions should inefficient units combine the inputs or outputs to become efficient.

A (non)linear program does estimation of the relative efficiencies. Charnes, Cooper, and Rhodes (CCR) proposed CCR (input-oriented) model assuming constant returns to scale, convexity, and free disposability. To linearize the fraction of efficiency in formula (2), to maximize its numerator keeping the denominators constant and to ensure the efficiency is greater than or equal to 1, respectively, we get the following program for the nonnegative weights $u_{\mu,j,t}$ and $v_{\mu,i,t}$ [5],

$$\begin{aligned} \sum_{j=1}^k u_{\mu,j,t} x_{\mu,j,t} &\rightarrow \max \quad \text{s. t.} \\ \sum_{i=1}^l v_{\mu,i,t} y_{\mu,i,t} &= 1 \\ \sum_{i=1}^l v_{\mu,i,t} y_{\mu,i,t} - \sum_{j=1}^k u_{\mu,j,t} x_{\mu,j,t} &\geq 0 \\ u_{\mu,j,t} &\geq 0 \\ v_{\mu,i,t} &\geq 0. \end{aligned} \quad (3)$$

Analogously, Banker, Charnes, and Cooper introduced DEA model called BCC [5], assuming variable returns to scale, convexity and free disposability, which is mathematically guaranteed by adding a constant κ to model (3),

$$\begin{aligned} \sum_{j=1}^k u_{\mu,j,t} x_{\mu,j,t} - \kappa &\rightarrow \max \quad \text{s. t.} \\ \sum_{i=1}^l v_{\mu,i,t} y_{\mu,i,t} &= 1 \\ \sum_{i=1}^l v_{\mu,i,t} y_{\mu,i,t} - \sum_{j=1}^k u_{\mu,j,t} x_{\mu,j,t} &\geq \kappa \\ u_{\mu,j,t} &\geq 0 \\ v_{\mu,i,t} &\geq 0. \end{aligned} \quad (4)$$

Finally, Deprins, Simar and Tulkens broke the assumptions of convexity and defined FDH model [5] enabling free disposability hull, mathematically done by a variable $\lambda_{\mu,j,t} \in \{0, 1\}$ in model (4),

$$\begin{aligned} \sum_{j=1}^k \lambda_{\mu,j,t} u_{\mu,j,t} x_{\mu,j,t} - \kappa &\rightarrow \max \quad \text{s. t.} \\ \sum_{i=1}^l v_{\mu,i,t} y_{\mu,i,t} &= 1 \\ \sum_{i=1}^l v_{\mu,i,t} y_{\mu,i,t} - \sum_{j=1}^k \lambda_{\mu,j,t} u_{\mu,j,t} x_{\mu,j,t} &\geq \kappa \\ u_{\mu,j,t} &\geq 0 \\ v_{\mu,i,t} &\geq 0 \\ \lambda_{\mu,j,t} &\in \{0, 1\}. \end{aligned} \quad (5)$$

All computations around the DEA models were performed using R language and statistical environment [8, 9].

C. Data envelopment analysis models connected in time series

Given the model of our choice, since we can estimate n efficiencies for each unit in each time point $t \in \{1, 2, \dots, T\}$ based on the criterial matrix C_t , finally, we finally get n time series for all n units (one time series $\{\phi_{\mu,1}, \phi_{\mu,2}, \dots, \phi_{\mu,T}\}$ per each unit μ). A similar approach based on DEA models and their connections in time series was published in [10]; however, authors built the time series for efficiencies rather for initial five weeks after the 100-th positive case in a given country.

III. RESULTS

Time developments of proportions of vaccinated people, number of hospitalized patients, and reproduction rates for the given period are in Fig. 1, Fig. 2, and Fig. 3., respectively. In Fig. 2, the numbers of hospitalized patients are below 200 per one million inhabitants except for Bulgaria (BGR), Latvia (LVA), and Estonia (EST).

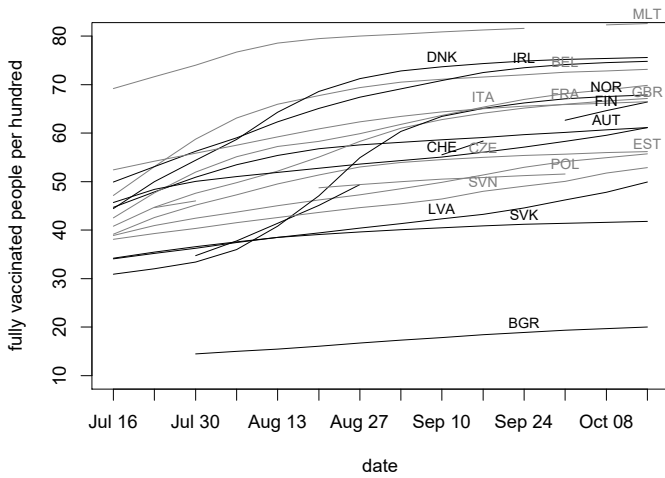


Fig. 1. Time development of proportions of fully vaccinated people per one hundred inhabitants for different countries in summer 2021 (colours of the lines and appropriate labels are matched but may mutually differ to make them better distinguishable; interrupted curves indicate missing values; abbreviations of the countries are explained in the text).

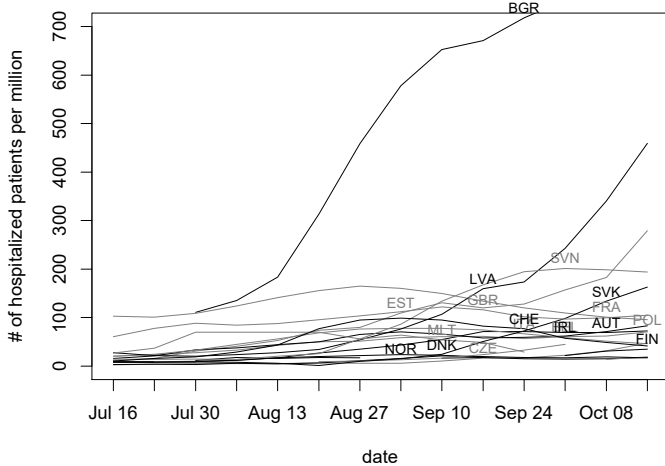


Fig. 2. Time development of hospitalized patients per one million inhabitants for different countries in summer 2021 (colours of the lines and appropriate labels are matched but may mutually differ to make them better distinguishable; abbreviations of the countries are explained in the text).

To respect the convexity and inner logic of DEA models, we used rather reciprocal values of the number of hospitalized patients and reproduction rate. Thus, the better the anti-pandemic country management is, the higher the reciprocal value of hospitalized patients and the reciprocal value of reproduction rate.

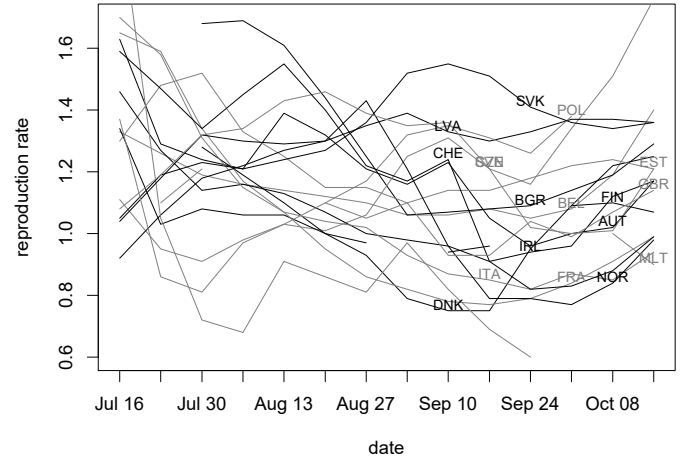


Fig. 3. Time development of reproduction rate for different countries in summer 2021 (colours of the lines and appropriate labels are matched but may mutually differ to make them better distinguishable; abbreviations of the countries are explained in the text).

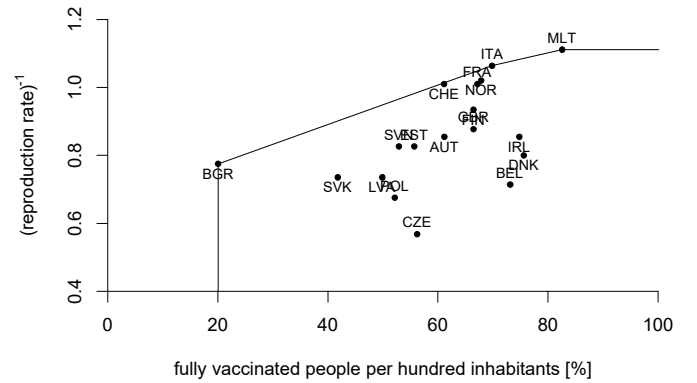


Fig. 4. A relation between the reciprocal reproduction rate and the proportion of fully vaccinated people using BCC model for October 15, 2021 (abbreviations of the countries are explained in the text).

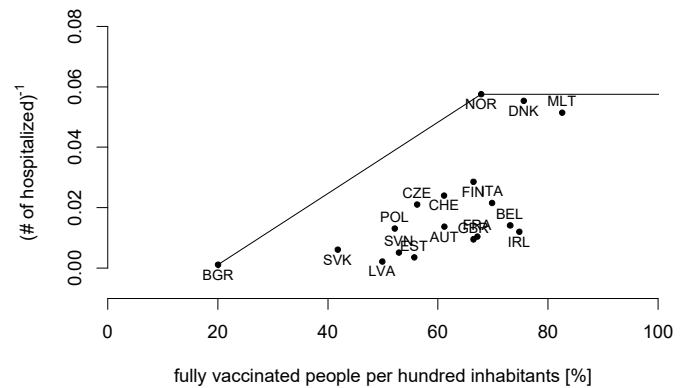


Fig. 5. A relation between the reciprocal number of hospitalized patients and the proportion of fully vaccinated people using BCC model for October 15, 2021 (abbreviations of the countries are explained in the text).

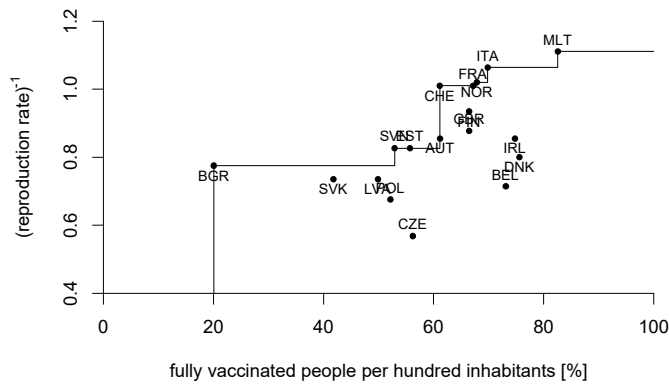


Fig. 6. A relation between the reciprocal reproduction rate and the proportion of fully vaccinated people using FDH model for October 15, 2021 (abbreviations of the countries are explained in the text).

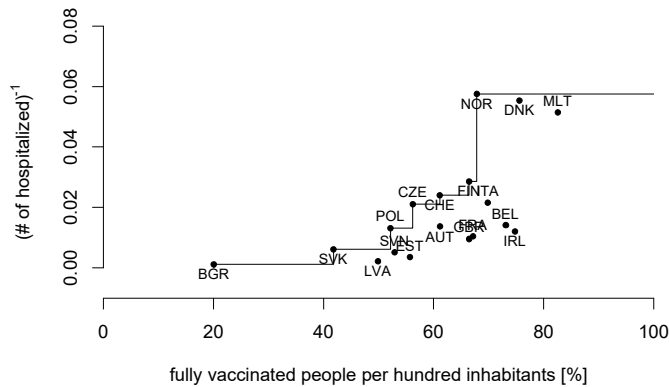


Fig. 7. A relation between the reciprocal number of hospitalized patients and the proportion of fully vaccinated people using FDH model for October 15, 2021 (abbreviations of the countries are explained in the text).

For a time point of October 15, 2021, there is a relation between the reciprocal reproduction rate and the proportion of fully vaccinated people by BCC and FDH model in Fig. 4 and Fig. 6, respectively, and a relation between the reciprocal number of hospitalized patients and the proportion of fully vaccinated people by BCC and FDH model in Fig. 5 and Fig. 7, respectively. Therefore, units, i. e. countries, that are located on the production frontiers are effective ones according to the models.

Similarly to the previous figures, for a time point of October 15, 2021, the relative efficiencies of all units (countries) estimated by different models are shown in Table 1. As we can see, the CCR model, since based on constant returns to scale, may identify only a few units as effective (constant, linear production frontiers may go through only plot origin $[0, 0]$ and one other unit's coordinates); i. e. in our case, only Bulgaria and Norway seem to be effective. On the other hand, the FDH model usually identifies many units as effective since its production frontiers are "staircase"-shaped (see Fig. 6 and 7); i. e. more than 50 % of all units identified as effective. Thus, the BCC model seems to be the most feasible. According to BCC models, regardless of whether input- or output-oriented, Bulgaria (BGR), Italy (ITA), Malta (MLT), Norway (NOR), and Switzerland (CHE) are identified as technically efficient in the defined anti-pandemic management. This is roughly in correspondence with

Fig. 4 and 5; however, while the models are multidimensional, the plots are only two-dimensional and, thus, plots' results are only informative.

TABLE I. RELATIVE EFFICIENCIES OF DIFFERENT DEA MODELS (CCR, BCC AND FDH), BOTH INPUT-ORIENTED ("IN") AND OUTPUT-ORIENTED ("OUT") FOR OCTOBER 15, 2021.

country	CCR		BCC		FDH	
	in	out	in	out	in	out
Austria (AUT)	0.51	1.95	0.57	1.19	1.00	1.18
Belgium (BEL)	0.38	2.60	0.42	1.51	0.84	1.49
Bulgaria (BGR)	1.00	1.00	1.00	1.00	1.00	1.00
Czechia (CZE)	0.53	1.90	0.66	1.70	1.00	1.00
Denmark (DNK)	0.86	1.16	0.87	1.04	0.90	1.04
Estonia (EST)	0.41	2.41	0.52	1.19	0.95	1.00
Finland (FIN)	0.67	1.50	0.75	1.11	1.00	1.00
France (FRA)	0.49	2.05	0.90	1.04	0.91	1.00
Ireland (IRL)	0.40	2.49	0.46	1.27	0.82	1.24
Italy (ITA)	0.61	1.65	1.00	1.00	1.00	1.00
Latvia (LVA)	0.40	2.52	0.42	1.29	0.84	1.00
Malta (MLT)	0.79	1.26	1.00	1.00	1.00	1.00
Norway (NOR)	1.00	1.00	1.00	1.00	1.00	1.00
Poland (POL)	0.51	1.97	0.58	1.42	1.00	1.00
Slovakia (SVK)	0.54	1.84	0.58	1.23	1.00	1.00
Slovenia (SVN)	0.46	2.18	0.55	1.17	1.00	1.00
Switzerland (CHE)	0.70	1.43	1.00	1.00	1.00	1.00
United Kingdom (GBR)	0.43	2.31	0.57	1.19	0.92	1.15

Finally, in Fig. 8 and 9, there are time series for relative efficiencies coming from the BCC input- and output-oriented models, respectively. We can see the relative distances and shapes are roughly constant in time for all countries; thus, that being said, it may indicate the relative efficiencies for countries are roughly correlated. However, the efficiencies in time are hardly stable (they are non-stationary). There seems to be no country with an efficiency equal to 1 in the long term period. Still, some countries, namely Bulgaria (BGR), Denmark (DNK), Czechia (CZE), and perhaps Switzerland (CHE), keep their efficiencies close to 1 for a longer period.

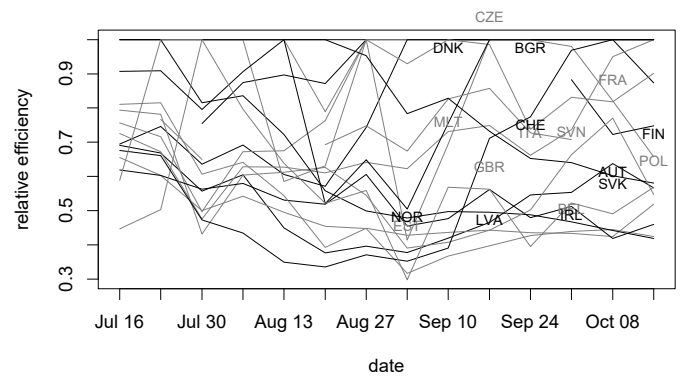


Fig. 8. Time series of relative efficiencies based on BCC input-oriented model (colours of the lines and appropriate labels are matched but may mutually differ to make them better distinguishable; interrupted curves indicate missing values; abbreviations of the countries are explained in the text).

For comparison, we also show time series for relative efficiencies estimated by FDH model, both by input- and output-oriented, in Fig. 10 and 11, respectively. As we could see both in Fig. 6 and 7 for FDH models limited by the time point of

October 15, 2021, and in Fig. 10 and 11, the FDH models tend to overestimate a final number of efficient units, i. e., countries.

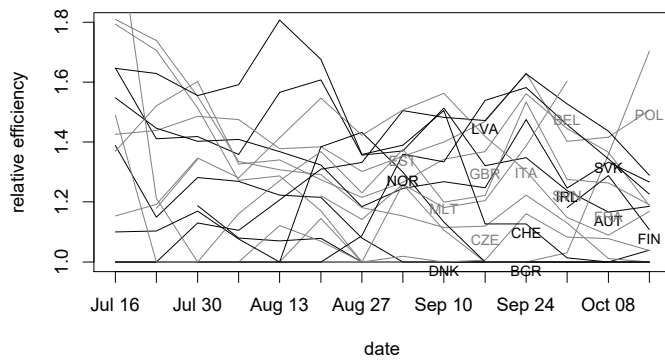


Fig. 9. Time series of relative efficiencies based on BCC output-oriented model (colours of the lines and appropriate labels are matched but may mutually differ to make them better distinguishable; interrupted curves indicate missing values; abbreviations of the countries are explained in the text).

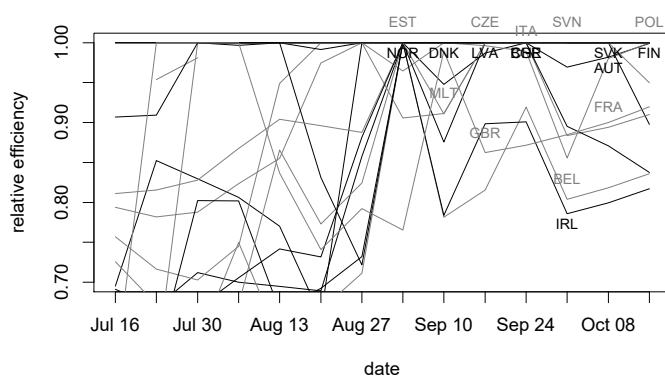


Fig. 10. Time series of relative efficiencies based on FDH input-oriented model (colours of the lines and appropriate labels are matched but may mutually differ to make them better distinguishable; interrupted curves indicate missing values; abbreviations of the countries are explained in the text).

The role of Bulgaria (BGR) as an efficient country is likely, besides others, caused by the fact that the country is at a level of roughly 20 % of fully vaccinated inhabitants, the only one in our dataset; thus, it has got no “competitor” and is efficient despite its low level of vaccination and increasing COVID-19 reproduction rate.

IV. CONCLUSIONS

DEA models enable to estimate of relative efficiencies in COVID-19 anti-pandemic country-level management. Considering the proportions of fully vaccinated people, countries as independent units may variously control whether usual hygienic rules are followed by their inhabitants or may motivate them to get anti-COVID-19 vaccination. The better the anti-pandemic by a given country is, the lower the number of hospitalized patients and reproduction rate is. Thus, the effectiveness of the anti-pandemic management can be quantified by relative efficiencies coming from DEA models. Using data for the period of summer 2021 and the DEA models, particularly the BCC model, and appropriate visualizations, we identified several countries with efficiencies equal to or close

to 1. Those may serve as good examples for others – other countries may consider adopting some aspects of their anti-pandemic country policy to become effective, too. Furthermore, we tried to build time series using DEA-based efficiencies; however, those time series seem not to be stationary in time.

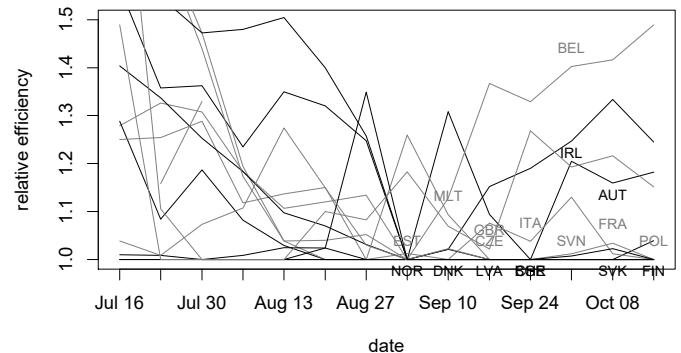


Fig. 11. Time series of relative efficiencies based on FDH output-oriented model (colours of the lines and appropriate labels are matched but may mutually differ to make them better distinguishable; interrupted curves indicate missing values; abbreviations of the countries are explained in the text).

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